

Chapter 1

Using Visual Studio Code for Machine Learning Experiments

1.1 Introduction

Before performing Machine Learning experiments, it is necessary to have a suitable software environment for writing, executing, debugging, and testing Python programs. Although Python programs can be written using a simple text editor, modern software development is usually carried out using an Integrated Development Environment (IDE).

An IDE provides facilities such as code editing, syntax highlighting, debugging, automatic code completion, terminal access, and project management. These facilities help programmers write programs more efficiently and reduce errors.

Visual Studio Code (VS Code) is one of the most popular IDEs used for Python programming and Machine Learning. It is a free and open-source editor developed by Microsoft and is widely used by students, researchers, and industry professionals.

VS Code supports numerous programming languages and can be extended using plugins called extensions. By installing the Python extension, VS Code becomes a powerful environment for Machine Learning development.

This chapter explains how to install and configure Visual Studio Code for Machine Learning experiments using Python.

1.2 What is Visual Studio Code?

Visual Studio Code, commonly known as VS Code, is a lightweight yet powerful source code editor developed by Microsoft.

VS Code provides several useful features including:

- Syntax highlighting
- Automatic code completion
- Error detection
- Integrated terminal
- Debugging support

- Git integration
- Jupyter notebook support
- Extension support

Because of these features, VS Code has become one of the most widely used tools for Machine Learning and Data Science projects.

1.3 Installing Python

Python must be installed before VS Code can be used for Machine Learning.

1.3.1 Windows

1. Visit:

<https://www.python.org/downloads/>

2. Download the latest stable version.
3. Run the installer.
4. Select the option
Add Python to PATH
before clicking Install.
5. Complete the installation process.

1.3.2 Ubuntu

Open a terminal and execute:

```
sudo apt update  
sudo apt install python3 python3-pip
```

1.4 Verifying Python Installation

After installation, verify that Python has been installed correctly.

1.4.1 Windows

Open Command Prompt and type:

```
python --version
```

1.4.2 Ubuntu

Open a terminal and type:

```
python3 --version
```

If Python is installed successfully, a version number similar to the following will appear.

```
Python 3.12.0
```

1.5 Installing VS Code

1.5.1 Windows

1. Download VS Code from:
<https://code.visualstudio.com/>
2. Run the installer.
3. Accept the license agreement.
4. Select
Add to PATH
during installation.
5. Complete the installation.

1.5.2 Ubuntu

Open a terminal and execute:

```
sudo apt update  
sudo apt install code
```

If the package is unavailable:

1. Download the .deb package.
2. Install using:

```
sudo dpkg -i code*.deb  
sudo apt -f install
```

1.6 Launching VS Code

After installation:

1.6.1 Windows

Open:

Start Menu → Visual Studio Code

1.6.2 Ubuntu

Type:

<code>code</code>

in the terminal.

1.7 Understanding the VS Code Interface

When VS Code starts, several components become visible.

- Activity Bar
- Explorer
- Search Window
- Source Control
- Extensions View
- Editor Window
- Terminal Window
- Status Bar

The Explorer is used to manage project files.

The Editor Window is used to write programs.

The Terminal is used to execute commands.

The Extensions View is used to install additional functionality.

1.8 Installing the Python Extension

VS Code requires the Python extension to support Python development.

1. Open VS Code.
2. Press:
`Ctrl + Shift + X`
3. Search for:
Python
4. Select the extension published by Microsoft.
5. Click Install.

After installation, VS Code automatically provides Python syntax highlighting, code completion, debugging support, and execution support.

1.9 Creating a Python Project

A project is a collection of related files.

To create a project:

1. Create a folder.
2. Open VS Code.
3. Select:

File → Open Folder

4. Choose the folder.

The project folder now appears in the Explorer panel.

1.10 Creating and Running the First Python Program

Create a file named:

hello.py

Enter:

```
print("Hello Machine Learning")
```

Save the file.

Run it using:

```
python hello.py
```

Output:

Hello Machine Learning

1.11 Using the Integrated Terminal

VS Code provides an integrated terminal.

Open it using:

Terminal → New Terminal

or

Ctrl + `

The integrated terminal eliminates the need to switch between applications.

1.12 Installing Machine Learning Packages

Machine Learning experiments require additional libraries.

1.12.1 Windows

```
pip install numpy pandas matplotlib scikit-learn
```

1.12.2 Ubuntu

```
pip3 install numpy pandas matplotlib scikit-learn
```

1.13 Verifying Package Installation

Create a file named:

testlibraries.py

```
import numpy
import pandas
import matplotlib
import sklearn

print("Libraries Installed Successfully")
```

Output:

Libraries Installed Successfully

1.14 Running Machine Learning Experiments

A typical Machine Learning experiment consists of:

1. Loading data.
2. Preprocessing data.
3. Selecting a model.
4. Training the model.
5. Testing the model.
6. Evaluating the results.
7. Visualizing the results.

All these steps can be performed inside VS Code.

1.15 Using Jupyter Notebooks in VS Code

Many Machine Learning practitioners use Jupyter notebooks.

To enable notebook support:

1. Open Extensions.

2. Search for:

Jupyter

3. Install the extension.

Create a notebook using:

File → New File → Jupyter Notebook

Notebook files use the extension:

`.ipynb`

Jupyter notebooks allow code execution one cell at a time and are widely used for Machine Learning experiments.

1.16 Debugging Python Programs

Debugging helps identify program errors.

To debug:

1. Click beside a line number to create a breakpoint.

2. Press F5.

3. Execute the program.

4. Observe variable values and execution flow.

Debugging is particularly useful when Machine Learning programs produce unexpected results.

1.17 Common Errors and Their Solutions

1.17.1 Python Not Found

Error:

```
python is not recognized
```

Solution:

Reinstall Python and ensure that

Add Python to PATH

is selected.

1.17.2 Module Not Found

Error:

`ModuleNotFoundError`

Solution:

Install the missing package using `pip`.

1.17.3 Incorrect Python Interpreter

Sometimes VS Code selects the wrong interpreter.

Select the correct interpreter using:

Ctrl + Shift + P

and choose:

Python: Select Interpreter

1.18 Advantages of VS Code for Machine Learning

- Free and open source.
- Cross-platform.
- Integrated terminal.
- Excellent Python support.
- Jupyter notebook integration.
- Powerful debugging facilities.
- Extensive extension ecosystem.
- Suitable for both beginners and advanced users.

1.19 Summary

Visual Studio Code provides a complete environment for Machine Learning development. After installing Python, VS Code, the Python extension, and the required Machine Learning libraries, students can create, execute, debug, and visualize Machine Learning programs efficiently. Because of its simplicity, flexibility, and rich feature set, VS Code has become one of the most popular platforms for Machine Learning and Data Science projects.